



**Combined CPS Energy Reactive Power
Planning Project & CPS Energy Helotes
345/138-kV Switching Station and
Autotransformer Addition at Eastside
Switching Station Project – ERCOT
Independent Review Scope**

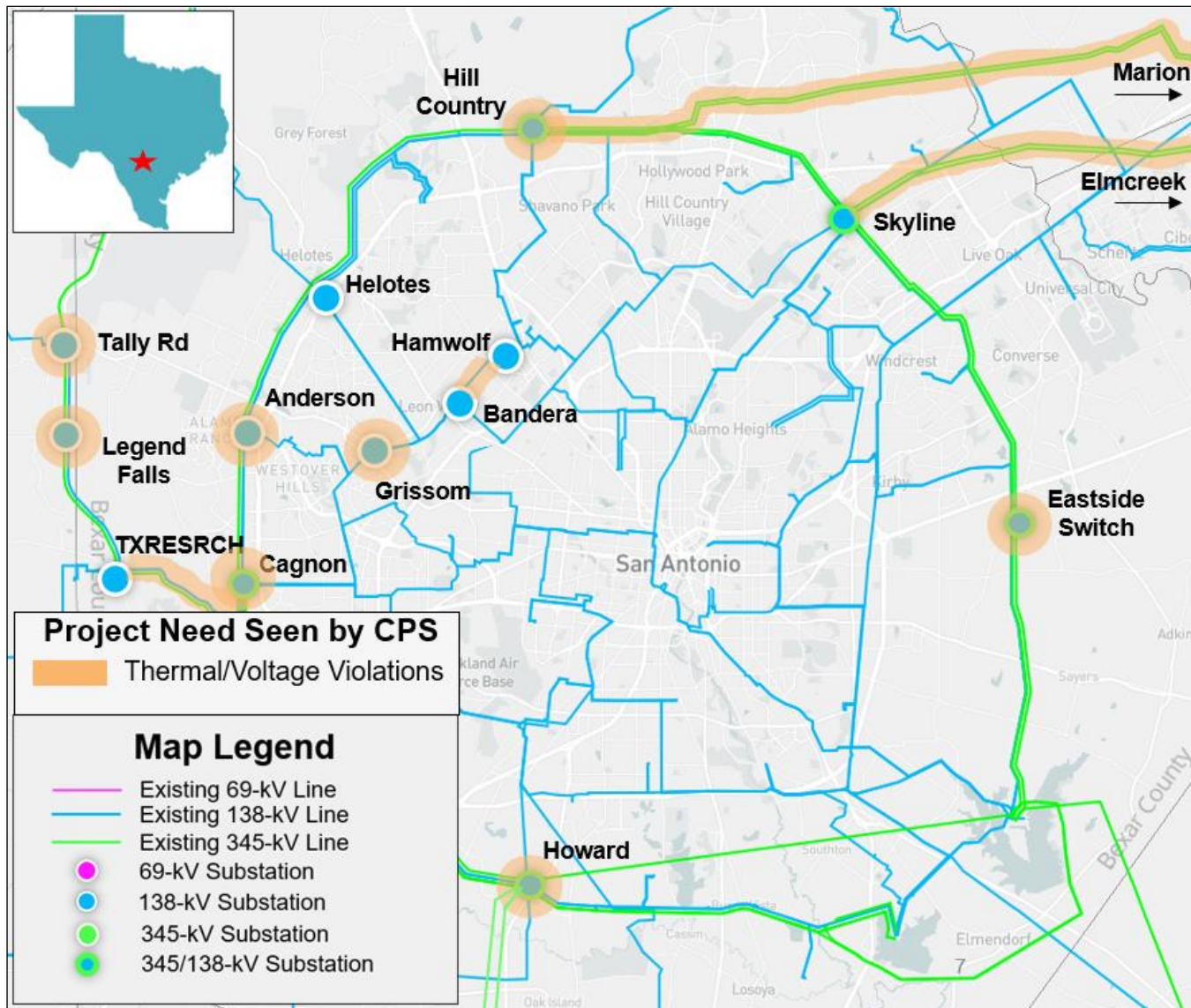
Travis Head

RPG Meeting
July 29, 2025

Introduction

- CPS Energy Reactive Power Planning Project (25RPG013) for Regional Planning Group (RPG) review in May 2025
 - This is a Tier 1 project with an estimated cost of \$116.50 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated in-service date (ISD) is December 2029
 - This project is needed to address post-contingency voltage violations and improve voltage recovery performance in the Bexar County
- CPS Energy Helotes 345/138-kV Switching Station and Autotransformer Addition at Eastside Switching Station Project (25RPG017) for Regional Planning Group (RPG) review in May 2025
 - This is a Tier 1 project with an estimated cost of \$110.0 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated ISD is Summer 2028 and Summer 2029
 - This project is needed to address post-contingency thermal violations in the Bexar County
- ERCOT is currently conducting a single ERCOT Independent Review (EIR) by combining these two projects (25RPG013 and 25RPG017)

Study Area Map with Project Needs Seen by CPS



Combined Proposed Project by CPS

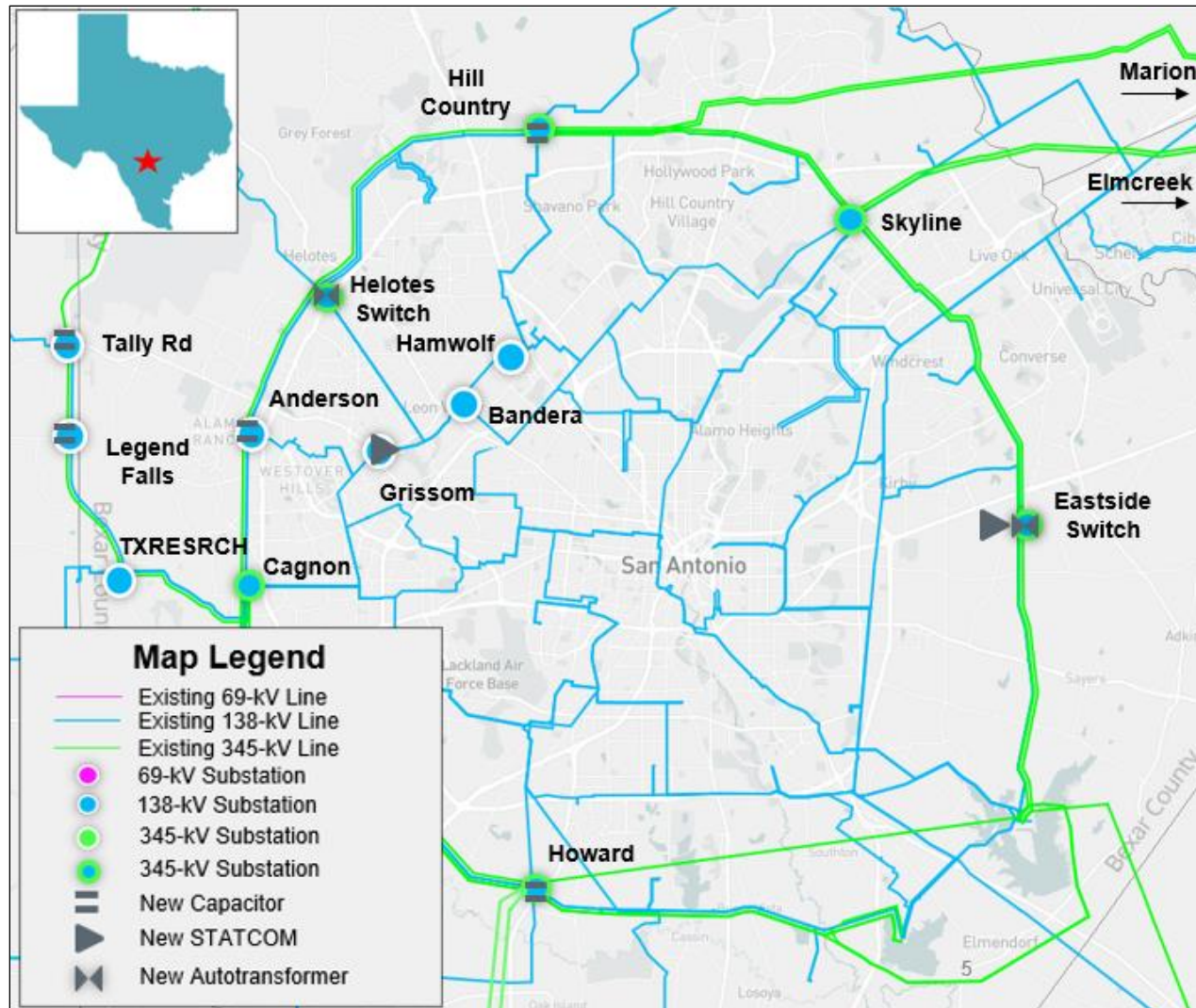
- Install the following sized capacitors at the following substations:
 - 50 MVAR, Hill Country
 - 50 MVAR, Anderson
 - 25 MVAR, Legend Falls
 - 50 MVAR, Howard
 - 25 MVAR, Talley Rd
- Install the following STATCOMs at the following substations:
 - 300 MVAR, Grissom
 - 300 MVAR, Eastside Switch

Combined Proposed Project by CPS

cont'd

- Construct a new Helotes 345-kV substation;
- Convert the existing Helotes 138-kV substation to a new 345/138-kV switching substation;
- Install two new 345/138-kV autotransformers at the new Helotes 345/138-kV switching substation, normal and emergency ratings of at least 600 MVA;
- Loop in the existing Hill Country to Cagnon 345-kV transmission line into the new Helotes 345-kV substation; and
- Install a new 345/138-kV autotransformer at the existing Eastside 345/138-kV substation, normal and emergency ratings of at least 600 MVA

Map with combined Proposed Project by CPS



Study Assumptions – Base Case

- Study Area
 - Bexar County in South Central Weather Zone is considered for this study
 - Atascosa, Bandera, Comal, Guadalupe, Kendall, Medina, and Wilson Counties which are electrically close to project location
- Steady-State Base Case
 - Final 2024 Regional Transmission Planning (RTP) 2030 summer peak load case published on Market Information System (MIS) in December 2024, will be updated to construct the study base case
 - Case: 2024RTP_2030_SUM_12222024
 - Link: <https://mis.ercot.com/secure/data-products/grid/regional-planning>

Study Assumptions – Transmission

- Based on the February 2025 Transmission Project and Information Tracking (TPIT) posted on the ERCOT website projects within the study area with in-service dates before June 2030 within the study area will be added to the study base case if not already modeled in the case
 - TPIT Link: <https://www.ercot.com/gridinfo/planning>
 - See Appendix A for the list of transmission projects added to the case
- Transmission projects identified in the 2024 RTP in the study area that have not been approved by RPG will be removed from the study base case
 - See Appendix B for the list of transmission projects removed from the case

Study Assumptions – Generation

- New generation that met the Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before June 2030 in the study area at the time of the study will be added to the case if not already modeled in the RTP case based on June 2025 Generator Interconnection Status (GIS) report published in MIS in May 2025
 - Link: <https://www.ercot.com/gridinfo/resource>
 - See Appendix C for the list of generation projects added to the case
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2024 RTP Final cases

Study Assumptions – Load & Reserve

- Loads in study area
 - Load level in the study area will be maintained consistent with the Final RTP cases
- Reserve
 - The reserve will be maintained consistent with the 2024 RTP

Contingencies & Criteria

- Contingencies
 - NERC TPL-001-5.1 and ERCOT Planning Criteria
 - (<http://www.ercot.com/mktrules/guides/planning/current>)
 - P0 (System Intact)
 - P1, P2-1, P7 (N-1 conditions)
 - P2-2, P2-3, P4, and P5 (345-kV only)
 - P3 (G-1+N-1: G-1 of Leon Creek U1, San Miguel U1, Guadalupe Energy Center CTG 1, and JK Spruce U1)
 - P6-2 (X-1+N-1: X-1 of Cagnon X1, Howard Road X1, and Hill Country X1 345/138-kV transformers)
- Criteria
 - Monitor all 69-kV and above busses, transmission lines, and transformers in the study region (excluding generator step-up transformers)
 - Thermal
 - Use Rate A for normal conditions
 - Use Rate B for emergency conditions
 - Voltage
 - Voltages exceeding their pre-contingency and post-contingency limits
 - Voltage deviations exceeding 8% on non-radial load buses

Study Procedure

- Need Analysis
 - The reliability analysis will be performed to identify the need to serve the projected area load using the study base case
- Project Evaluation
 - Project alternatives will be tested to satisfy the NERC and ERCOT reliability requirements
 - ERCOT may also perform the following studies
 - Planned maintenance outage
 - Long-Term Load-Serving Capability Assessment
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Subsynchronous Resonance (SSR) Assessment
 - TSP(s) will provide Cost and Feasibility Assessment
- Congestion Analysis
 - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Deliverables

- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final recommendation – Q3 2025

Thank you!



Stakeholder comments also welcomed through:

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Appendix A – Transmission Projects Added

TPIT/RPG No	Project Name	Tier	Project ISD	TSP	County(s)
89912	GVEC_Olmos to Wilson 138 kV TL, T297	Tier 4	12/01/26	GVEC	Guadelupe

Appendix B – Transmission Projects Removed

RTP Project ID	Project Name	TSP(s)	County(s)
2024-SC19	Hillje (44200) to Zorn (7042) 345-kV Line Upgrades	CNP, AEN, LCRA TSC	Wharton, Fayette, Bastrop, Caldwell, Guadalupe
2024-SC26	Shaula (5380) to Elm Creek (5133) to Cachena (5068) 345-kV Line Upgrades	CPS	Wilson, Guadalupe, Dewitt

Appendix C – Generation Added

GINR	Project Name	Fuel	Project COD	Capacity (MW)	County
23INR0078	Shaw Solar	Solar	04/29/2026	124.65	Bandera
23INR0479	Taormina Storage	BESS	05/26/2029	231.86	Bexar
24INR0533	Padua Grid BESS Unit 2	BESS	03/15/2026	150.94	Bexar
28INR0024	Padua Grid BESS Unit 3	BESS	05/15/2026	201.35	Bexar